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Value co-creation practices in the service of innovative digital products and their effects on consumer eudaimonic well-being

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ABSTRACT

Innovative digital products (IDPs) are novel products that combine traditional physical components with smart (AI) and connectivity (IoT) components. While these innovative technological products are designed to enhance consumer well-being, their actual effects remain controversial. We propose resolving the well-being paradox in smart services through consumer-driven value co-creation with IDPs. However, how consumers proactively leverage IDPs to co-create value and enhance their well-being remains underexplored. Our research aims to identify the specific value co-creation practices between consumers and IDPs that can improve consumer eudaimonic well-being and examine the underlying mechanisms. In Study 1, we employed practice theory—a sociological lens—to qualitatively identify three types of consumer-IDP value co-creation practices: (1) adapting practices, (2) crafting practices, (3) connecting practices. In Study 2, we quantitatively demonstrated that these practices positively affected basic need satisfaction, which in turn enhanced consumer eudaimonic well-being. Our research provides theoretical contributions and managerial implications for facilitating consumer-driven value co-creation and improving consumer well-being in IDP services.

1. Introduction

Innovative digital products (IDPs) are novel products enabled by digital technologies, which combine traditional physical components with smart (AI) and connectivity (IoT) components (Belingheri & Neirotti, 2019; Pesch et al., 2021). A prominent example of IDPs is smart home (SH) devices (Wiesböck & Hess, 2020), such as smart lights, locks, speakers etc.—all of which are smart and connected. These products have rapidly expanded into various aspects of daily life. Latest data predicts that global shipments of SH devices will reach 931 million units by 2025 (IDC, 2025), signaling a booming market prospect.

The introduction of new technologies often brings both expected and unexpected consequences in users' environments (Beaudry & Pinsonneault, 2005). While these innovative technological products are designed to enhance consumer well-being (Attié & Meyer-Waarden, 2022), their actual effects remain controversial. Some studies show that AI products can perform daily tasks on behalf of consumers and offer personalized recommendations. This offers significant convenience and empowerment, thereby promoting consumer well-being (Kang & Shao, 2023; Shen et al., 2025). In contrast, others argue that delegating tasks to AI may erode consumers' control and decision-making autonomy, ultimately harming their well-being (Gonçalves et al., 2024; Puntoni et al., 2020). The key to enhancing well-being lies in the innovative interaction possibilities enabling users to co-create value with smart products. Yet, these AI-powered interactions can also undermine consumers' autonomy and self-efficacy. Thus, it is crucial to understand how consumers can enhance their well-being while engaging in value co-creation with products in smart services.

The current research addresses the well-being paradox in smart services through the lens of consumer-driven value co-creation with IDPs. This perspective aligns with the service research priority of "customer proactivity for well-being" proposed by Ostrom et al. (2021). The priority emphasizes the growing expectation for consumers to take a more active role in managing their well-being through digital technologies (Tikkanen et al., 2023). Existing research also highlights that well-being is closely linked to individuals' agentic participation in goal pursuit (e.g., personal growth) (Huta & Ryan, 2010; Ryan & Deci, 2001). Through active engagement, consumers can benefit from innovative value co-creation with digital technology while maintaining their control and decision-making autonomy. Therefore, identifying

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consumer-driven value co-creation activities with IDPs can offer critical insights into resolving the ongoing debate about smart products' impact on consumer well-being.

Importantly, IDPs hold significant potential for consumer-driven value co-creation, yet this area remains underexplored. Existing research has primarily focused on value co-creation between consumers and independent AI products (smart components), such as voice Q&A and personalized recommendations through direct interactions (Knote et al., 2021). AI products typically empower consumers in a one-sided manner, leaving little room for deeper consumer participation. In contrast, IDPs mark a pivotal shift in value co-creation. The connectivity components of IDPs substantially increase product malleability, creating greater opportunities for consumer-driven value co-creation. On one hand, these components enable IDPs to gather information and receive commands from both users and other connected devices. On the other hand, they allow consumers to proactively reprogram and modify product functions via seamless synchronization among connected IDPs (Novak & Hoffman, 2019). For instance, consumers can program lights to automatically turn on at sunset, turn off at midnight, and adjust their colors to match different moods (Hoffman & Novak, 2018). Despite these advantages, there has been limited research on how consumers proactively engage with IDPs to co-create value in daily lives. This gap not only limits our understanding of value co-creation in smart services, but also obscures the relationship between smart products and consumer well-being. Failure to address this issue could result in a disconnect between technological advancements and human well-being.

Motivated by existing research gaps, our research question is: *what specific value co-creation practices between consumers and IDPs can improve consumer eudaimonic well-being and how.* We focus on eudaimonic well-being, defined as human functioning and self-realization (Ryff & Singer, 2008). While hedonic well-being is important, eudaimonic well-being better reflects contemporary societal expectations that individuals actively manage their own well-being. More crucially, it aligns with our focus on consumer-driven value co-creation practices. Additionally, a fully functioning individual is better positioned to achieve hedonic well-being (Tikkanen et al., 2023). We employed a mixed-method approach to address the research question. Study 1 used qualitative methods to systematically identify value co-creation practices between consumer and IDPs: adapting practices, crafting practices and connecting practices. By introducing practice theory as a theoretical lens, we posit that value co-creation emerges from the interaction between material elements (IDPs) and consumer elements (competence and meaning), rather than material properties alone (Shove et al., 2012). Study 2 quantitatively tested how these practices impacted eudaimonic well-being through the mechanism of basic need satisfaction (autonomy, competence, and relatedness), based on self-determination theory (Deci & Ryan, 2000; Ryan et al., 2008). Meanwhile, we compared the different effects across identified value co-creation practices. Using structural equation model, Study 2 validated these theoretically derived pathways.

This research makes important contributions to the literature. First, we advance consumer well-being research by exploring how consumers actively manage their well-being through digital technologies. This offers new insights into resolving the well-being paradox in smart services, and responds to the service research priority of "customer proactivity for well-being". Second, we expand value co-creation research in smart services by shifting focus to IDP contexts. Unlike previous studies that mainly focus on consumers' passive role in value co-creation with independent AI products (Knote et al., 2021; Mele et al., 2022), we illustrate how IDPs with connectivity components empower consumers' dominant role in value co-creation, revealing a consumer-driven model. Our research also contributes to ongoing discussions on social (non-economic) value creation in the information systems (IS) field. Finally, we extend practice theory literature by fully considering the agency of materials. Our model demonstrates how IDPs' technological properties and human agency jointly shape value co-creation practices. Practically, our findings offer helpful guidance for managers seeking to enhance consumer well-being and facilitate value co-creation in IDP services. We also provide actionable insights for consumers on how to proactively manage their well-being through digital technologies.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature. Section 3 outlines the research design of the mixed-method approach employed. Sections 4 and 5 present the details of Studies 1 and 2, respectively. Section 6 discusses the findings, theoretical and practical contributions, as well as the limitations and future research directions. Finally, we conclude with a summary of the research findings.

2. Theoretical background

2.1. Smart products and consumer well-being

Well-being has long been recognized as a crucial outcome in IS and marketing research (Anderson et al., 2013; Attié & Meyer-Waarden, 2022). Mainstream research typically differentiates between two key conceptualizations of well-being: hedonic and eudaimonic (Rahmani et al., 2018; Ryan & Deci, 2001). Hedonic well-being is primarily characterized by the presence of positive emotions and the absence of negative ones, such as pleasure, fun, and enjoyment (Ryan & Deci, 2001). In contrast, eudaimonic well-being is associated with a sense of meaning and purpose in life. It is broadly interpreted as the experience of being a fully functioning individual (Ryan & Deci, 2001), such as self-actualization, personal growth, and a sense of mastery in life (Huta, 2013).

Although both hedonic and eudaimonic well-being are important in technology interactions, our research focuses on eudaimonic well-being for three key reasons. First, contemporary societal and technological transformations have elevated the importance of eudaimonic well-being (Tikkanen et al., 2023). Consumers are increasingly expected to assume greater responsibility to manage their own well-being, and the emphasis on "customer proactivity for well-being" aligns closely with eudaimonic well-being (Ostrom et al., 2021). Second, consumer-driven value co-creation with IDPs is inherently tied to eudaimonic well-being. By proactive participating in value co-creation, consumers experience self-actualization, improved quality of life, enhanced mastery, freedom, and a sense of fulfillment—all resonating with eudaimonic well-being (Anderson et al., 2013; Ryff & Singer, 2008). Third, while hedonic well-being is often associated with short-term pleasure, eudaimonic well-being offers deeper, more enduring satisfaction. It not only encompasses hedonic elements but also extends beyond them to foster a more meaningful and lasting sense of well-being (Deci & Ryan, 2008; Ryan et al., 2008).

The impact of smart products on consumer well-being has attracted considerable academic interest, yet it remains a topic of debate. AI technology enhances task efficiency and convenience by taking over routine activities and providing ubiquitous service access (Shen et al., 2025), which in turn promotes well-being (Kang & Shao, 2023). For instance, using SH devices to manage household chores can decrease the domestic workload and energy costs. It enhances consumer well-being by creating a more comfortable and convenient living environment (Sequeiros et al., 2022). Additionally, AI-powered information retrieval systems (e.g., personalized recommendations, smart Q&A) help consumers to quickly access high-quality, relevant information, simplifying decision-making and reducing cognitive search costs, thereby enhancing well-being (Wang et al., 2025).

However, delegating tasks to smart products may also diminish users' control and decision-making autonomy, which harm their well-being (Gonçalves et al., 2024; Raff et al., 2024). Spencer (2023) argues that technological advancement can trigger replacement anxiety while gradually eroding personal autonomy. Mehmood et al. (2024) suggest that smart products make consumers feel manipulated, which restricts their perceived freedom in decision-making. Puntoni et al. (2020) contend that outsourcing tasks to smart products can reduce consumers' self-efficacy and heightened concerns on being replaced. Collectively, these studies underscore that smart products may inadvertently impact overall well-being.

We propose that consumer-driven value co-creation serves as the key pathway to resolve the well-being paradox in smart services. This process involves consumers' active engagement in innovative interactions with products to co-create value, enabling users to simultaneously benefit from technological convenience while maintaining control and autonomy. This perspective aligns with the service research priority of "customer proactivity for well-being" (Ostrom et al., 2021). It emphasizes that consumers should play an active role in managing their well-being. Existing research also supports that eudaimonic well-being more fundamentally derives from individuals' agentic participation in goal pursuit (e.g., personal growth) (Huta & Ryan, 2010; Ryan & Deci, 2001). For instance, Henkens et al. (2021) found that smart products engagement boosts well-being by improving self-efficacy and reducing technology anxiety. Zhang et al. (2022) demonstrate that proactive customer participation enhances digital skills and further increase well-being.

In summary, we contend that the consumer-driven value co-creation model is essential for promoting consumer eudaimonic well-being in smart services. Next, we will discuss the value co-creation between consumers and IDPs.

2.2. Value co-creation between consumers and IDPs

Value co-creation refers to the benefits derived from integrating resources through activities and interactions with collaborators within a customer's service network (McColl-Kennedy et al., 2012). Recent scholarship has extended this concept to smart services, exploring value co-creation between humans and innovative technical objects (Knote et al., 2021). Existing research mainly examines value co-creation with independent AI products (smart components) through direct consumer interactions, focusing on dimensions like intelligent Q&A, task execution, and optimized decision-making (see Table 1). These interactions have demonstrated significant benefits, including improved information access, enhanced service experiences, and increased satisfaction and well-being (Kot & Leszczyński, 2022; Manser Payne et al., 2021; Mele et al., 2021; Peltier et al., 2024). However, they also have potential drawbacks. This form of value co-creation is still largely unilateral, with smart products exert agency while consumers remain relatively passive. For instance, relying on AI for decision-making can erode consumers' task control and decision-making autonomy (Gonçalves et al., 2024; Puntoni et al., 2020). This constrains our understanding of consumer-driven value co-creation in smart services.

In contrast, IDPs offer great potential for consumer-driven value co-creation. IDPs refer to new products embodied in digital technology, consisting of physical, smart, and connectivity components (Belingheri & Neirotti, 2019; Pesch et al., 2021). The connectivity components of IDPs substantially increase product malleability, offering more opportunities for consumers to actively shape their interactions with products. On one hand, these components expand the scope of products' information collection and command receipt. Unlike independent AI products, IDPs can collect information and receive commands from both users and other connected IDPs. On the other hand, these components enable consumers to more easily reprogram and modify product functionality (Novak & Hoffman, 2019). These advantages allow consumers to adjust the output of multiple devices in real time to deliver personalized service experiences. Thus, the connectivity components significantly empower consumers' dominant role in value co-creation and establish a consumer-driven model.

However, existing research on IDPs has largely focused on how organizations develop and enhance the performance of these products (Hendler, 2021), as well as the factors influencing consumer adoption or resistance (Hubert et al., 2025; Koohang et al., 2022; Mamonov & Koufaris, 2020; Stieglitz et al., 2023). Few studies have examined value

Table 1
Relevant literature on value co-creation in smart services.

Research perspective	Research content	Characteristics	Related literature
The value co-creation between consumers and smart components	**Dimensions of value co-creation:** ● Intelligent Q&A (e.g., communication and chat) ● Task execution (e.g., performing specific tasks on behalf of users) ● Optimized decision-making (e.g., content recommendations, behavior suggestions, providing information to assist decision-making)	Products unidirectionally empowers consumers through consumer-independent AI interactions, with consumers playing a relatively passive role in value co-creation.	Barile et al., (2024); Chandra & Rahman, (2024); Knote et al., (2021); Kot & Leszczyński, (2022); Mele et al., (2021); Mele et al., (2022)
	Consumer outcomes of value co-creation: ● Information value, customer experiences, behavioral intentions, satisfaction, well-being, and viability		Kot & Leszczyński, (2022); Manser Payne et al., (2021); Mele et al., (2021); Peltier et al., (2024)
The value co-creation between consumers and connectivity components	**Dimensions of value co-creation:** ● Monitor and remote control	Only simple value co-creation dimensions are identified, without highlighting the contribution of connectivity components to the consumers' dominant role and to consumer-driven value co-creation	Gonçalves & Patrício, (2022)
	Consumer outcomes of value co-creation: ● Utilitarian value, hedonic value and transformative value; customers' continuance intentions and word-of-mouth		Balaji & Roy, (2017); Harvey et al., (2020)
	Dimensions of value co-creation: ● *Adapting practices, crafting practices, connecting practices*	*Connecting components empower consumers to reprogram and modify the use of connected IDPs, facilitating consumer-driven value co-creation, which significantly enhances consumer eudaimonic well-being*	*Our current research*
	Consumer outcomes of value co-creation: ● *Eudaimonic well-being*		

co-creation between consumers and IDPs (see Table 1). Harvey et al. (2020) explore how consumers craft SH products to co-create value from a value-outcome perspective. Gonçalves and Patrício (2022) explore value co-creation in smart energy services, focusing on how consumers achieve energy efficiency and savings through monitoring and remotely controlling products. While these studies addressed consumer-IDP value co-creation, they neither fully explore this specific theme nor properly differentiate it from value co-creation with independent AI products. Thus, there is a research gap regarding how consumers actually use IDPs to co-create value. This gap not only limits our understanding of value co-creation in smart services, but also hinders the resolution of the ongoing debate on the impacts of smart products on consumer well-being. Our research aims to apply practice theory to systematically identify specific value co-creation activities when consumers fully utilize the smart and connectivity components of IDPs.

2.3. The practice-based perspective on value co-creation

Social practice theory, or simply practice theory, provides a distinctive explanation of the social world and how it changes (Bourdieu, 1977; Giddens, 1984). Practices refer to the routinized way (s) in which bodies are moved, objects are handled, subjects are treated, things are described and the world is understood (Reckwitz, 2002). In line with practice theory, practices are the ontological unit, shifting the analytical focus from individual agency and social structure to the practices themselves (Reckwitz, 2002; Shove et al., 2012).

Practices are organized by various interacting elements (Schatzki, 1997). Our work aligns with a recent and well-accepted proposal that breaks practices down into materials, competences and meanings (Shove et al., 2012; Thomas & Epp, 2019). Specifically, materials refer to things, technologies, tangible physical entities, and the stuff from which objects are made; competences refer to knowledge and skills necessary for practices; meanings refer to symbolic meanings, ideas, aspirations and purposes (Shove et al., 2012). Practices emerge, persist, evolve and disappear through the formation and dissolution of links among these interacting elements (Shove et al., 2012; Thomas & Epp, 2019). Moreover, as elements combine to create recognizable practices, practices can be linked to other practices to form nexuses among interacting practices (Shove et al., 2012). Thus, changes in a particular practice require not only a reconfiguration of its constituent elements, but also the re-establishment of connections with related practices.

Practice theory is well-suited for studying consumer-IDP value co-creation. Traditional technological theories, such as technological determinism, assume that the technology will be built as planned and used in specific ways to achieve anticipated outcomes. However, the true value of technology lies not in the artifact itself, but in its application within specific contexts to accomplish tasks (Kempton, 2022). The operation and outcomes of technologies are not fixed or predetermined; they emerge over time through interactions with humans in practice (Feldman & Orlikowski, 2011). According to practice theory, we posit that value co-creation between consumers and IDPs depends on the interaction between material elements (IDPs) and consumer elements (competence and meaning), rather than merely the material properties of the technology. The key is that the innovative technology must align with consumer elements and truly integrate into consumers' daily practices. Prior research has argued that technology is not inherently valuable or meaningful; it only gains these attributes when people actually engage with it in practice (Feldman & Orlikowski, 2011; Shove et al., 2012). Furthermore, practice theory has also been widely used in the studies of value co-creation across various contexts (Kelleher et al., 2019; McColl-Kennedy et al., 2012; Schau et al., 2009). Therefore, we use practice theory as a theoretical lens to understand value co-creation between consumers and IDPs.

2.4. Self-determination theory

To identify the mechanisms through which value co-creation practices between consumers and IDPs affect consumer eudaimonic well-being, we draw on self-determination theory (SDT). SDT posits that humans have three universal psychological needs: autonomy, competence, and relatedness (Deci & Ryan, 2000; Ryan et al., 2008). The need for autonomy refers to the sense of volition, the experience of choice, and feeling like the initiator of one's own actions. The need for competence involves feeling effective in one's life and capable of attaining desired outcomes. The need for relatedness refers to the desire to feel connected to others and to cultivate positive social relations. These basic psychological needs are regarded as essential nutriments for personal growth, motivation, and well-being (Ryan & Deci, 2001).

SDT offers a solid theoretical foundation for this research. First, the consumer-IDP value co-creation is a consumer-driven process. When consumers actively dedicate time and effort to redesigning and tailoring IDPs according to their personal needs and household contexts, the proactive engagement enhances their sense of control, self-efficacy, and family harmony. These outcomes align perfectly with SDT's core proposition that individuals flourish when they create conditions that satisfy their basic psychological needs (Ryan & Deci, 2000). Second, SDT is widely recognized as a theory explaining eudaimonic well-being (Rahmani et al., 2018; Ryan et al., 2008). This theory positions eudaimonia as central to well-being, and elaborates the pathways to self-realization through basic psychological need satisfaction (Ryan & Deci, 2001). This perspective resonates strongly with eudaimonic principles of actualizing potential and pursuing meaningful goals (Ryan et al., 2008). Both SDT and eudaimonic well-being prioritize self-actualization and intrinsic growth rather than external pleasure (i.e., hedonic well-being) (Ryan et al., 2008). Consequently, we argue that SDT offers an ideal framework for understanding how consumer-IDP value co-creation practices impact eudaimonic well-being through need-fulfilling experiences.

3. Research design

To achieve our research objectives, we employed a mixed-method research design with two studies. In Study 1, we applied qualitative methods to identify specific value co-creation practices between consumers and IDPs. In Study 2, we developed our conceptual model and hypotheses based on the identified practices and relevant literature. We then used structural equation model to empirically test these proposed relationships.

4. Study 1 Identifying value co-creation practices between consumers and IDPs

4.1. Overview

Study 1 employed discovery-oriented qualitative methods to identify consumer-IDP value co-creation practices in daily life. An inductive approach is particularly appropriate, as the phenomena of value co-creation practices in the context of IDP services is not well understood in existing literature. This method involves the iterative collection and analysis of field data and literature to gain deeper insights into understudied phenomena (Chen et al., 2024; Strauss & Corbin, 1998).

Additionally, we selected SH as our research setting for several key reasons. First, SH encompasses a wide range of IDPs across various categories, rather than representing a single IDP category. Second, many bestselling IDPs currently fall under the SH category, such as smart voice assistants and smart locks (Duggal, 2024). Third, the boundaries of SH are not clearly defined, as non-home products, such as wearable technology and in-car smart devices, can also be connected to in-home devices. Lastly, the literature frequently cites SH as typical examples of IDPs (Alvarado-Vargas et al., 2020; Wiesböck & Hess, 2020).

4.2. Method

4.2.1. Data collection

Consistent with prior exploratory research (Johnson & Sohi, 2016), we applied a theoretical sampling plan to select samples and collect data. Theoretical sampling is a non-random method designed to identify samples that can provide deeper insights into the studied phenomenon (Strauss & Corbin, 1998). Our full dataset included in-depth semi-structured interviews, participant observations and additional online records (see Web Appendix A). A summary of data collection is presented in Table 2.

We conducted in-depth interviews with each respondent to gain a fine-grained and comprehensive understanding of value co-creation practices between consumers and IDPs. Interview data was the primary focus of our study. Additionally, to develop a more embodied and

Table 2
Summary of the data collection.

Data source	Content
Semi-structured interview	We intentionally sought consumers of SH from various age groups, skill levels, work backgrounds, and family structures. In total, we recruited 26 participants, each receiving a financial payment of 50 CNY. To gain a more comprehensive understanding of consumers' value co-creation practices with SH, we also found five practitioners from the SH industry who actively used these products in daily lives. We conducted in-depth face-to-face or phone interviews with each respondent, with durations ranging from 30 min to over 2 h. After obtaining consent, each interview was audiotaped, and the audio data was transcribed within 24 h. For some interviewees, we followed up via phone or WeChat to clarify uncertain information.
Participant observation	The first author installed and learned to use SH, encouraging his family to integrate these products into their lives. The author participated in and observed the family's practices extensively. Additionally, the author visited each of the five interviewees in their homes three times, engaging in informal conversations while observing their value co-creation practices with SH.
Additional online records	The authors viewed over 400 videos from two content creators on TikTok and Bilibili who specialize in SH devices, each with a significant following. The authors collected consumer reviews and posts from platforms like "What is Worth Buying" and "Zhihu," searching for SH-related keywords about usage, service, experiences, and practices.

material understanding of these practices (Godfrey et al., 2022), we employed participant observation. Specifically, the first author installed SH and encouraged the families to integrate them into daily lives, while actively participating in and observing their use. We also visited the homes of five interviewees on three separate occasions, observing their usage and engaging in informal conversations. Finally, to identify patterns and themes that might not have emerged from interviews and observations alone, we further searched for online records related to SH usage, primarily for supplementary and confirmatory purposes. By utilizing multiple data sources, we aimed to achieve triangulation.

4.2.2. Data analysis

We applied practice theory as a preliminary analytical framework to guide the categorization of our data. The widely recognized three-element model—comprising materials, competences, and meanings—offers a foundational lens for analyzing practices. From a practice-based perspective, practices emerge, evolve, or dissolve through the dynamic interplay among these elements. In this study, the coding process primarily focused on how the material element (i.e., the novel capabilities of SH devices) smoothly interacts with consumer-related elements (i.e., competence and meaning) to facilitate value co-creation.

Our data analysis employed coding techniques from grounded theory, following the established scheme consisting of open, axial, and selective coding (Chen et al., 2024; Strauss & Corbin, 1998). First, during the open coding phase, we coded the data line-by-line to identify relevant concepts as zero-order categories, adhering closely to informants' own language. In the axial coding phase, we contextualized these zero-order categories with relevant literature and engaged in a constant comparative process—iteratively moving between the emerging data themes and theoretical insights on practice elements and their interrelations. We then reassembled the descriptive zero-order categories into more theoretically abstract first-order categories. Finally, in the selective coding phase, we distilled the emergent first-order categories into second-order categories. This stage involved refining the theory to identify central themes. See Table 3 for coding results.

We followed the method of prior qualitative studies to maintain analytical rigor and establish the trustworthiness of the data (Johnson & Sohi, 2016). First, we applied constant comparison among the data, memos, and literature to validate the emergent categories and findings. Also, we shared our findings with five practitioners in the SH industry, inviting them to objectively evaluate the accuracy and effectiveness of our results and provide feedback. We further refined and improved the coding results during this discussion process. Second, we sought three new participants for interviews, achieving theoretical saturation as no new categories or properties emerged from the additional data collection.

Table 3
Zero-order, first-order, and second-order coding categories.

Zero-order categories	First-order categories	Second-order categories
·Commanding devices by voice ·Questioning and answering by voice	Adapting to direct interaction capabilities	Adapting practices
·Indirectly controlling devices by voices ·Indirectly controlling devices by smart ways ·Indirectly interacting with family by devices	Adapting to indirect interaction capabilities	
·Proactively mastering new knowledge and skill ·Redesigning and experimenting with creative usage methods	Proactively investing	Crafting practices
·Modifying technology usages ·Modifying task usages	Creatively reproducing device usages	
·Connecting tasks by smart scene linkages ·Connecting tasks by smart responses ·Connecting tasks by remote management	Connecting convergent practices	Connecting practices
·Sharing ·Quantifying life activities ·Orchestrating family affairs	Connecting generative practices	

4.3. Findings

Our findings reveal that value co-creation practices between consumers and SH consist of three parts: (1) adapting practices, (2) crafting practices, and (3) connecting practices. Each type of practice involves a complex interplay of core elements—namely, materials, competences, and meanings—within a practice or among related practices. The core of value co-creation practices lies in aligning the newly introduced materials of IDPs with consumers' competences and meanings. In adapting practices, the focus is on how consumers adapt IDPs to fulfill relatively traditional meanings. Crafting practices emphasize how consumers creatively modify the regular use of IDPs to achieve new meanings. Connecting practices highlight how consumers use IDPs to facilitate connections between different practices, either through the IDPs themselves or through their derived meanings.

4.3.1. Adapting practices

SH devices, as newly introduced material elements, integrate advanced smart and connectivity components that differentiate them from traditional home devices. These novel components enhance the interactive capabilities of IDPs, requiring consumers to develop competencies to utilize devices' potential. This adaptation process underpins the first type of value co-creation practice we identify: adapting practices. These practices involve consumers reconfiguring newly introduced SH elements with their competences to achieve meaningful outcomes. The key aspect of these practices is that consumers must adapt to the novel interactive capabilities of SH devices. Specifically, two types of adapting practices were identified: (1) adapting to direct interaction capabilities, and (2) adapting to indirect interaction capabilities. Through these novel interaction processes, consumers actively co-create value with SH devices.

(1) Adapting to direct interaction capabilities. With the integration of smart components (AI), SH devices can autonomously process and respond to consumer commands. As consumers adapt to the devices' communicative capabilities, they can realize intended meanings through direct interactions with these innovative products.

There are two primary types of direct interactions: "commanding devices by voice" and "questioning and answering (Q&A) by voice." "Commanding devices by voice" refers to SH devices understanding consumer voice commands and executing the corresponding tasks. For example, many participants mentioned that they directly controlled smart TVs through voice commands, such as adjusting volume or switching channels. "Q&A by voice" involves SH devices engaging in real-time communication with consumers, such as answering questions about the weather, and even having conversations. These new interactive methods significantly reduce the common inconveniences in traditional home settings, such as locating remote controls, walking to fixed switches, and using phones to search for information.

(2) Adapting to indirect interaction capabilities. Enabled by connectivity components (IoT), SH devices can communicate with one another, allowing consumers to interact with one SH device indirectly through another as an intermediary. By adapting to the connectivity components, users can efficiently complete daily tasks by indirect interactions with innovative technology.

There are three primary types of indirect interactions: "indirectly controlling devices by voice," "indirectly controlling devices by smart methods," and "indirectly interacting with family by devices." "Indirectly controlling devices by voice" is similar to the "voice command" method mentioned earlier. Yet, the key difference lies in how the voice commands are processed. Instead of the target device responding directly, the process is mediated through another device. This involves two main steps: first, the consumer issues a voice command to an intermediary device, which then relays the command to the target device. Next, the target device responds to the consumer's voice command. As one participant said "*I typically control a smart airer using voice commands via a smart speaker (C7).*"

In addition to common voice commands, consumers also indirectly control devices by smart methods. For example, they can set trigger conditions using door or window sensors, and issue commands through smart devices such as wireless switches or magic squares. One uploader shared: "*I installed a door sensor on the wardrobe and mounted a cabinet light at the top to address the difficulty of finding clothes in dim lighting. When the wardrobe door opens, the cabinet light automatically turns on, making it easier to find clothes (O).*"

"Indirectly interacting with family by devices" focuses on how consumers use SH devices as intermediaries to interact with family members, offering convenience that was previously unavailable. For instance, many participants highlighted the positive role of smart door locks in ensuring family safety. One participant mentioned, "*I installed a smart door lock and a peephole, which gives me peace of mind about the safety of my parents and children when they are home alone. I can monitor visitors at any time (C9).*" Additionally, many participants shared how they use smart cameras and phones to remotely monitor elderly or ill family members while away at work. This capability proves especially valuable in easing anxieties about their loved ones' welfare during absences.

4.3.2. Crafting practices

Adapting practices primarily focus on how consumers adapt to the new interactive capabilities of SH devices to realize the expected and routine meanings. However, we found that as consumers become more familiar with SH devices, they begin to actively explore more sophisticated and innovative using ways. By leveraging the smart and connectivity components, they craft unique usage patterns tailored to their personal needs and tastes (Harvey et al., 2020).

Building on Campbell's (2005) concept of crafting consumption, we identified a second set of value co-creation practices: crafting practices. Crafting consumption refers to activities in which individuals both design and make the products that they consume. In this process, consumers proactively apply their skills and knowledge to modify standardized offerings (Brock & Johnson, 2021). This notion aligns with value co-creation practices in SH context. Beyond adapting practices, consumers actively redesign and creatively modify their use of SH. Through these practices, they not only develop improved crafting competence in transforming devices' material capabilities, but also often discover new, unanticipated meanings. This aligns with craft consumption theory, which positions creative self-expression as craft consumers' primary motivation. Our analysis delineates two core dimensions of crafting practices: "proactively investing" and "creatively reproducing device usages."

(1) Proactively investing. To develop crafting competence, consumers proactively invest efforts in learning and studying how to use SH technologies more deeply. "Proactively investing" encompasses two key aspects: "proactively mastering new knowledge and skills" and "redesigning and experimenting with creative usage methods." First, users need to dedicate efforts to master new knowledge and skills. They often proactively learn and study the settings and usage of SH devices through various online resources or consulting friends. Then, they continuously apply their knowledge, hands-on skills and imagination to redesign and experiment with creative usage methods. As one participant said, "*I follow several content creators on Bilibili and usually watch their videos to learn how to reset SH devices. I've experimented with many of their recommended methods to explore creative using ways for my unique needs (C17).*" Although these crafting processes often demand considerable effort, many informants reported intrinsic enjoyment from the creative challenge.

(2) Creatively reproducing device usages. In the crafting process, consumers play the role of "prosumers"—both producers and consumers of product usage (Toffler, 1980). The core of crafting practices lies in consumers creatively reproduce the nonstandard usage of SH devices. "Creatively reproducing device usages" is manifested in 2 key aspects: "modifying technology usages" and "modifying task usages". Modifying technology usages refers to consumers reprogramming or reconfiguring device operations to perform familiar tasks in creative ways. The emphasis here is on unconventional technological applications, rather than using devices for entirely new tasks. As the online data showed: "*The smart electric fan lacks intelligent functionality, since we can't change its gear by voice command or smart ways. So, I devoted a lot of time to developing an alternative control method using a dual-key wireless switch, which overcame the limitation and allowed smart control of the fan's settings* (O)."

Additionally, consumers often repurpose devices to accomplish new tasks. "Modifying task usages" focuses on modifying device use for unintended or unexpected tasks, uses that these devices were not originally designed to fulfill. As a participant shared, "*Initially, I used Xiao Ai (a smart speaker) for basic functions like weather updates and alarms. Then one evening, too tired to watch TV, I played Chinese idiom games with it. Over time, Xiao Ai became my nighttime companion—telling stories and communicating with me. I customized its voice to sound like my favorite male celebrity, which made living alone feel less intimidating, especially when I heard strange noises at night or felt lonely (18).*"

Creatively reproducing device usages often allows consumers to seek self-expression that transcends the conventionally functional meaning of a product. Many standardized use cases frequently fail to meet consumers' unique and dynamic needs. Thus, they seek to craft device usage to develop personalized solutions for their specific, non-standardized pain points or interests. For many users, this ongoing crafting process becomes a way to express their individuality. What begins as functional adaptation evolves into intrinsically rewarding practices, transforming mundane technology interactions into deeply meaningful experiences.

4.3.3. Connecting practices

Shove et al. (2012) argue that, just as elements are linked together to form recognizable practices, practices themselves can also be interconnected. As usage deepens, SH devices not only enhance individual single practice but also facilitate connections among them. An informant noted, *"It is still in the early stage of using SH, when you only control SH devices by voice command. The true value of SH emerges when you utilize smart linkages, executing several tasks simultaneously through one command, rather than just controlling individual devices (P1)."* We identified the third set of value co-creation practices as connecting practices. It means that using SH devices to link specific value co-creation practices with other practices.

Grounded in practice theory, practice connections are formed through shared elements. In our context, we found that connecting practices are mainly based on shared material (i.e., SH devices) and shared meaning (derived from device usage) elements. To theorize these connections, we draw upon the concepts of convergence and generativity from the digital innovation literature (Yoo et al., 2012). Convergent innovation refers to the integration of previously independent functions made possible by digital technologies. For instance, a 'triple-play' offering combines broadband internet, phone, and television services through the convergence of media content, storage, and distribution technologies. In contrast, generative innovation captures the ability of digital technologies to enable unanticipated, user-driven change, exemplified by the smartphone ecosystem that allows third-party developers to continuously reshape user experience. Applying the two concepts, we identified two forms of connecting practices: "connecting convergent practices" and "connecting generative practices." The former involves the integration of previously independent value co-creation practices through convergent usage of coordinated SH devices. The latter focuses on the connection of SH-enabled value co-creation practices with previously unrelated practices, driven by emergent and generative meanings.

(1) Connecting convergent practices. These practice connections mainly rely on SH devices themselves. Such device-based connections are largely rooted in the devices' original technical functions. Through connectivity components, multiple SH devices can operate in coordination. When one device performs a task, it can trigger other devices to act simultaneously. This interconnectedness enables consumers to effectively execute multiple tasks at once, linking previously independent practices. For example, a single voice command can synchronize several devices, eliminating the need to manually adjust each one using remote controls or fixed switches. By integrating the functions of different SH devices, consumers can merge separate tasks into cohesive operations, transforming formerly independent daily practices into a unified set of connected practices.

We identified three main types of connecting convergent practices: "connecting tasks by smart scene linkages," "connecting tasks by smart responses," and "connecting tasks by remote management." Smart scene linkages are widely regarded as one of the most valuable advantages of SH, enabling consumers to perform multiple tasks simultaneously. Consumers can realize scene linkages by multiple connected devices, which can be triggered to execute tasks automatically without manual intervention. This allows several consumer practices to be carried out at the same time. As a participant described setting up a "coming home" scene: "*I used to come home exhausted after work, and manually controlling the lights and the curtains felt like a burden. Now, simply unlocking my smart door triggers a seamless welcome routine: the lights automatically turn on, the curtains close, the TV and projector power up, and the rice cooker is already set to make congee. I can immediately settle onto the sofa to enjoy a meal and a movie, experiencing a deep sense of comfort and relaxation. What once required a series of manual steps now happens effortlessly the moment I walk through the door (C1).*" This case illustrates how practices can be linked to form bundles (Shove et al., 2012). The practices of turning the lights on/off, controlling the curtains, playing movies, and preparing meals are connected to form an integrated practice of "coming home."

Smart responses also effectively facilitate practice connections. SH devices can instantly process voice commands, allowing consumers to execute multiple tasks simultaneously through simple instructions. This functionality is particularly valuable in time-sensitive scenarios like cooking, as one participant described: *"I can easily prepare several dishes at once in the kitchen. I simply tell the smart oven what I want to cook; it automatically adjusts the settings, provides recipes, and answers my cooking questions. If someone comes to my home during that time, I can use voice commands to check who it is and unlock the door. Multitasking has become easy for me (C7)."*

Connecting convergent practices also extend beyond the home boundaries through long-finger control. Unlike "indirectly interacting with family by devices," this control focuses on managing devices while consumers are away from home. They can handle household tasks remotely by the connectivity of SH devices. One informant shared an example: *"Because I'm busy with work and often away from home, I used to feel disconnected from family life. But since installing SH, things have changed completely. I now feel much closer to my family, even when I'm not physically there. I use smart sockets to remotely control household devices, such as turning the air conditioning on/off. Later, I discovered that my child often used the computer secretly after school. To address this, I plugged it into a smart socket and used a mobile app to monitor its usage. When necessary, I remotely switch off the computer to help my child stay focused on studying (C14)."*

(2) Connecting generative practices. While connecting convergent practices are primarily based on shared elements of SH devices, the meanings behind these connections typically align with the devices' original, intended technical functions. In contrast, connecting generative practices are more rooted in the shared meaning element. The use of SH devices often generates new, unexpected meanings that go beyond their basic functionalities. These generative meanings can connect SH-enabled value co-creation practices with previously unrelated, or even seemingly contradictory, practices.

We mainly found three forms of generative practices that facilitate practice connections: "sharing," "quantifying life activities," and "orchestrating family affairs." "Sharing" is frequently mentioned by participants. With the growing usage of SH devices, consumers increasingly engage with them not only for functional benefits but also as a central topic of communication and social interaction. Many consumers actively share their SH usage experiences on social media, such as posting tutorials, exchanging tips with fellow enthusiasts, and showcasing smart lifestyles. This social sharing dimension is exemplified by one interviewee's account: "*Our community, called 'Xiaomi Whole-House Smart Groups,' brings together owners and enthusiasts who often exchange device recommendations and usage tips. Members often share their experiences with unique usage online and seek praise* (C2)." In offline settings, many informants also mentioned enjoying introducing their smart lifestyles to friends—often by demonstrating their SH usage during visits.

The ability to quantify life activities is a distinctive advantage of SH usage—one that traditional home devices cannot offer. Enabled by smart and connectivity components, SH devices allow comprehensive tracking of both personal behaviors and object usage, while aggregating data across multiple devices. This technological advancement has transformed SH devices from basic utilitarian tools into sophisticated platforms for scientific life management especially in areas like health and energy management. As one informant shared: "*I rely on SH devices to manage my health. I use a smart bracelet, smart weight scale, and smart bed to continuously measure and track key health metrics like heart rate, body composition, and sleep quality. All data syncs to my app, offering actionable insights to optimize my daily routines, diet, and exercise. Crucially, the instant alerts enable proactive health management by identifying potential issues before they become serious (C13).*"

The use of SH devices has also become a powerful way for consumers to orchestrate family affairs. These technologies enable users to elaborately coordinate device usage to address household needs and nurture relationships. Many participants emphasized their value in caring for family members. A participant said: *"Initially, I installed smart lights for* *convenience. But they took on new meaning when my wife began working late. I set the lights to automatically turn on from 10:30 PM to midnight, so she'd always return to a lit home. I added a bedside wireless switch, so she could turn off all lights without moving or speaking. She later said the lights seemed specially left for her and gave off a warm feeling (C3).*"

5. Study 2 Testing the effects of consumer-IDP value co-creation practices on eudaimonic well-being

5.1. Overview

Study 2 aimed to empirically examine the effects of identified value co-creation practices between consumers and IDPs on consumer eudaimonic well-being and the underlying mechanism. Based on SDT, we constructed a conceptual model linking value co-creation practices, eudaimonic well-being, and basic need satisfaction. Then, we used structural equation model to test the proposed hypotheses.

5.2. Conceptual model and hypotheses

We first developed the conceptual model based on the findings of value co-creation practices in Study 1 and literature review. In this model, we proposed that consumer-IDP value co-creation practices enhanced eudaimonic well-being through basic need satisfaction. Furthermore, we examined how different practices varied in their effects on both eudaimonic well-being and basic need satisfaction (see Fig. 1). Next, we present our hypotheses.

5.2.1. Consumer-IDP value co-creation practices and eudaimonic well-being

Eudaimonic well-being refers to the sense that one's activities are meaningful, contributing to a higher level of functioning and self-actualization (Deci & Ryan, 2008). It encompasses feelings of fulfillment, completeness, relationship and mastery in life (Huta, 2013). Consumer-IDP value co-creation practices can bring numerous benefits to consumers, such as control over their home environment, enhanced mastery skills, and harmonious family relationships. These outcomes help consumers realize their potential and foster self-growth, empowering them to pursue ideal lifestyles and enhance eudaimonic well-being.

Adapting practices can enhance consumer eudaimonic well-being. By allowing consumers to directly or indirectly control SH devices in flexible ways, these practices foster a sense of autonomy and reinforce perceived mastery over living environment. Such practices also facilitate indirect interactions with family members, helping to strengthen family bonds. As a result, adapting practices not only contribute to a more comfortable and convenient home life but also support a purposeful lifestyle aligned with consumers' personal values.

Compared to adapting practices, crafting practices are more effective in enhancing consumer eudaimonic well-being. Unlike simply adapting to predefined functions, crafting practices involve consumers actively investing effort to modify how they use SH devices. This proactive engagement fosters greater autonomy, supports skill development, encourages self-reflection, and enables the integration of new experiences (Ryan & Deci, 2001), leading to meaningful personal fulfillment. Moreover, when consumers tailor SH device usage for their families' benefit, the extra effort serves as a deeper expression of care and commitment (Garcia-Rada et al., 2022), thereby strengthening emotional connections within the family.

Compared to adapting practices, connecting practices are more effective in enhancing consumer eudaimonic well-being. While adapting practices typically involve using SH devices for single practice, connecting practices integrate multiple formerly independent routines into a coordinated practice assemblage. By leveraging SH devices to accomplish multiple tasks simultaneously, consumers gain greater control over home environment and experience enhanced autonomy. These empower them to manage complexity and optimize performance, promoting a sense of personal growth. Moreover, connecting practices not only improve household efficiency and reduce domestic conflicts but also enable consumers to orchestrate family activities using SH devices, fostering closer family relationships.

H1a. Adapting practices have a positive effect on eudaimonic well-being.

H1b. Crafting practices have a stronger positive effect on eudaimonic well-being than adapting practices.

H1c. Connecting practices have a stronger positive effect on eudaimonic well-being than adapting practices.

5.2.2. Consumer-IDP value co-creation practices and basic need satisfaction

We propose that consumer-IDP value co-creation practices can satisfy consumers' needs for autonomy, competence, and relatedness. Next, we present the detailed hypotheses.

Adapting practices enhance consumers' sense of autonomy. Various types of adaptive interactions allow consumers to volitionally choose how to control SH devices, such as voice command, smart ways or traditional switches. For instance, they can conveniently turn off the lights using voice commands. Moreover, SH devices offer more options for previously cumbersome tasks. Users once need to search for information using phones or computers, they can now choose to ask a smart speaker for answers such as the weather forecast.

Compared to adapting practices, crafting practices are more effective in enhancing consumers' autonomy. Adapting practices often rely on the preset functions of SH devices, offering consumers relatively limited opportunities for new choices and actions. In contrast, crafting practices empower them to dynamically redesign and modify the usage methods of SH devices based on their own preferences or needs. This approach

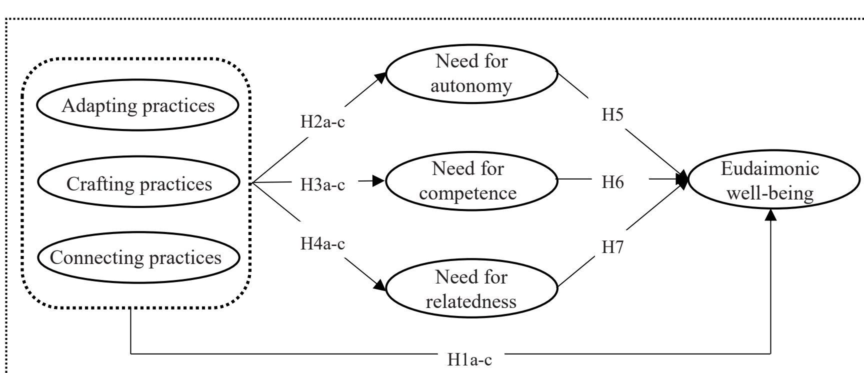

Fig. 1. Conceptual model.

provides greater freedom for creative device usage and task completion, thereby fostering a deeper sense of autonomy.

Compared to adapting practices, connecting practices are more effective in enhancing consumers' autonomy. While adapting practices are often carried out independently, device-based connecting practices enable consumers to coordinate multiple devices and perform several tasks simultaneously. These practices expand possibilities for multi-task execution, while also alleviating the burden of routine chores and granting them more freedom to pursue personal desires. Additionally, meaning-based connecting practices allow consumers to link previously unrelated activities, further reinforcing their autonomy through SH device usage.

H2a. Adapting practices have a positive effect on the need for autonomy.

H2b. Crafting practices have a stronger positive effect on the need for autonomy than adapting practices.

H2c. Connecting practices have a stronger positive effect on the need for autonomy than adapting practices.

Adapting practices also enhance consumers' sense of competence. These adaptive interactions make consumers' lives easier. They can control devices using voice commands or smart ways, eliminating the need for manual operations like flipping switches or searching for remote controls. Consumers acquire new capabilities to solve problems that are previously inconvenient or unachievable in a traditional home context.

Crafting practices enhance consumers' sense of competence more effectively than adapting practices. Compared with adaptive interactions with SH devices, crafting practices require consumers to proactively invest their knowledge, skills, and hands-on abilities to continuously experiment with and optimize SH device usage. This ongoing process involves accumulating knowledge, refining abilities, and overcoming challenges. Through such practices, consumers gradually develop stronger problem-solving skills, greater patience, and a sense of achievement, ultimately satisfying their competence needs more fully.

Connecting practices enhance consumers' sense of competence more effectively than adapting practices. By enabling the simultaneous completion of multiple tasks through SH devices, these practices greatly streamline household routines and improve task efficiency. This further improves consumers' competence in managing their detailed and complicated home environments. Moreover, connecting practices help consumers build competence in linking previously unrelated activities—often with personal goals that extend beyond basic device functions, such as quantifying life activities.

H3a. Adapting practices have a positive effect on the need for competence.

H3b. Crafting practices have a stronger positive effect on the need for competence than adapting practices.

H3c. Connecting practices have a stronger positive effect on the need for competence than adapting practices.

Adapting practices facilitate a sense of relatedness. These practices strengthen family bonds as consumers use SH devices to address household issues. Also, consumers can indirectly interact with family members by SH devices, such as using smart cameras to check on them and capture precious moments. Such practices help users stay connected with their families even when physically apart, conveying care and warmth within the family.

Crafting practices fulfill consumers' relatedness needs more effectively than adapting practices. Compared to adapting practices, crafting practices require consumers to invest greater efforts in proactively modifying SH device usage to better suit the preferences, needs, or routines of family members. These extra efforts convey deeper sincerity and love (Garcia-Rada et al., 2022). Additionally, crafting practices often involve mutual collaboration among family members to identify problems, propose improvements, and test solutions together. Such meaningful interactions reinforce mutual understanding and closeness.

Connecting practices fulfill consumers' relatedness needs more effectively than adapting practices. On one hand, device-based practice connections streamline household tasks, enhancing the overall efficiency of household operations. This multitasking integration reduces conflicts over daily chores and fosters a more harmonious home atmosphere. On the other hand, meaning-based practice connections allow consumers to orchestrate meaningful activities around family needs, such as setting lights for a late-returning family member. Such thoughtful arrangements deepen emotional bonds and nurture more intimate relationships.

H4a. Adapting practices have a positive effect on the need for relatedness.

H4b. Crafting practices have a stronger positive effect on the need for relatedness than adapting practices.

H4c. Connecting practices have a stronger positive effect on the need for relatedness than adapting practices.

5.2.3. Basic need satisfaction and eudaimonic well-being

SDT posits that satisfying individuals' basic needs is closely related to eudaimonic well-being (Rahmani et al., 2018; Ryan & Deci, 2001). Through consumer-driven value co-creation practices with IDPs, consumers develop new skills and enjoy greater freedom in product usage. These practices also enable consumers to better care for their families. Thus, these experiences yield meaningful outcomes, such as a sense of purpose in life, personal growth, positive relationships and environmental mastery. We propose that satisfying basic needs through value co-creation practices positively impacts eudaimonic well-being.

H5. Consumers' need for autonomy has a positive effect on eudaimonic well-being.

H6. Consumers' need for competence has a positive effect on eudaimonic well-being.

H7. Consumers' need for relatedness has a positive effect on eudaimonic well-being.

5.3. Method

5.3.1. Data collection

The survey consisted of two sections. The first section focused on measuring the variables explored in our study, while the second section gathered demographic information.

We recruited participants via the pre-screening function of Prolific, targeting SH device users. Before the formal survey, participants were first asked whether they had used SH devices. Only responses from actual SH device users were retained, resulting in a final valid sample of 400 participants. See Web Appendix B for participants' demographic information.

5.3.2. Measurement

In addition to the measurement of value co-creation practices, which were developed based on the findings from Study 1 and related literature (See Web Appendix C), all other measures were adopted from prior literature. All items were measured using a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree, see Web Appendix D).

5.4. Results

Structural equation model was employed, using AMOS 26, in our analysis.

5.4.1. Measurement model

We used Cronbach's α and confirmatory factor analysis (CFA) to evaluate the reliability and validity of the measurement model (see Table 4). All constructs met the reliability criteria, with all Cronbach's α values exceeding the acceptable threshold of 0.70 (Nunnally, 1978). We used standardized factor loadings (SFL), composite reliability (CR), and the average variance extracted (AVE) to verify the convergent validity. The SFL values should be higher than 0.6 (Comrey, 1973). Thus, we deleted the item of COP4, COP5, NFA2, NFR4, EW1, and EW3, and ran a second CFA with remaining items. As shown in Table 4, all SFL were greater than 0.6, and most exceeding 0.7, satisfying the recommended standard. The CR for all constructs were greater than 0.7 and the AVE for all constructs exceeded 0.5, which was considered excellent (Bagozzi & Yi, 1988). These results confirmed the good convergent validity of the measurement model. For discriminant validity, we compared the square root values of AVE and the construct correlations (Fornell & Larcker, 1981). As shown in Table 5, all square root values of AVE were greater than the correlations, supporting discriminant validity among the constructs.

We next estimated the fit of the model, which indicated that our data fit the measurement model well: $\chi^2 = 506.477$, df = 329, χ^2/df = 1.539 (suggested value < 5), CFI = 0.975 (suggested value > 0.9), TLI = 0.971 (suggested value > 0.9), RMSEA = 0.037 (suggested value < 0.8) (Hair et al., 2010).

5.4.2. Structural model

The structural model exhibited a good fit: $\chi^2 = 679.117$, df = 335, χ^2/df = 2.027, CFI = 0.951, TLI = 0.945, RMSEA = 0.051. Furthermore, the results indicated that all hypotheses were accepted.

Adapting practices positively influenced eudaimonic well-being, which refers to the total effect of adapting practices on eudaimonic well-being (H1a: $\beta = 0.124$, $p = 0.001$). Significantly positive relationships were also found between adapting practices and the needs for autonomy (H2a: $\beta = 0.134$, $p = 0.007$), competence (H3a: $\beta = 0.109$, $p = 0.001$), and relatedness (H4a: $\beta = 0.101$, $p = 0.005$), providing empirical support for H1a, 2a, 3a, 4a. We also found positive relationships between crafting practices and eudaimonic well-being ($\beta = 0.284$, $p < 0.001$), the needs for autonomy ($\beta = 0.427$, $p < 0.001$), competence ($\beta = 0.318$, $p < 0.001$), and relatedness ($\beta = 0.295$, $p < 0.001$). Parameter comparisons showed that crafting practices had significantly stronger effects on these outcomes than adapting practices (path coefficient differences: H1b: $\beta\_diff = 0.159$, $p = 0.010$; H2b: $\beta\_diff = 0.293$, $p = 0.006$; H3b: $\beta\_diff = 0.209$, $p = 0.015$; H4b: $\beta\_diff = 0.194$, $p = 0.012$), supporting H1b, 2b, 3b, 4b. Similarly, the positive effects of connecting practices on eudaimonic well-being ($\beta = 0.277$, $p < 0.001$), the needs for autonomy ($\beta = 0.377$, $p < 0.001$), competence ($\beta = 0.272$, $p < 0.001$), and relatedness ($\beta = 0.246$, $p < 0.001$) were significantly stronger than those of adapting practices (path coefficient differences: H1c: $\beta\_diff = 0.153$, $p = 0.010$; H2c: $\beta\_diff = 0.244$, $p = 0.023$; H3c: $\beta\_diff = 0.163$, $p = 0.046$; H4c: $\beta\_diff = 0.145$, $p = 0.044$), supporting H1c, 2c, 3c, 4c. Furthermore, the results indicated that basic need satisfaction had a positive effect on eudaimonic well-being. Specifically, we found significantly positive relationships between the needs for autonomy (H5: $\beta = 0.153$, $p < 0.001$), competence (H6: $\beta = 0.165$, $p < 0.001$), and relatedness (H7: $\beta = 0.241$, $p < 0.001$) and eudaimonic well-being, supporting H5-7.

We also tested whether basic need satisfaction mediated the effects of consumer-IDP value co-creation practices on eudaimonic well-being using the non-parametric bootstrapping regression technique with 5000 bootstrap samples. The bootstrap test was considered statistically significant (at 0.05 level) when the confidence interval (CI) did not include zero, indicating a significant indirect effect and established mediation (Hayes, 2022; Zhang et al., 2024). The results showed that three consumer-IDP value co-creation practices had indirect effects on eudaimonic well-being, through basic need satisfaction (see Table 6). Thus, the mediating effects of the basic need satisfaction were confirmed.

Table 4
The measurement model statistics.

Variable	Item	SFL	α	CR	AVE
Adapting Practices	AP1	0.697	0.787	0.897	0.687
	AP2	0.890			
	AP3	0.874			
	AP4	0.840			
Crafting Practices	CRP1	0.730	0.867	0.869	0.570
	CRP2	0.776			
	CRP3	0.747			
	CRP4	0.821			
	CRP5	0.695			
Connecting practices	COP1	0.826	0.834	0.838	0.564
	COP2	0.680			
	COP3	0.759			
	COP6	0.733			
Need for Autonomy	NFA1	0.879	0.927	0.928	0.763
	NFA3	0.880			
	NFA4	0.878			
	NFA5	0.856			
Need for Competence	NFC1	0.748	0.849	0.849	0.585
	NFC2	0.757			
	NFC3	0.769			
	NFC4	0.784			
Need for Relatedness	NFR1	0.735	0.821	0.824	0.610
	NFR2	0.821			
	NFR3	0.784			
Eudaimonic well-being	EW2	0.819	0.894	0.898	0.689
	EW4	0.812			
	EW5	0.868			
	EW6	0.819			

6. General discussion

Our research aims to explore *what specific value co-creation practices between consumers and IDPs can improve consumer eudaimonic well-being and how*. To address this, we employed a mixed-method approach. In Study 1, we used qualitative methods to identify three styles of value co-creation practices between consumers and IDPs. The first set of practices, "adapting practices," involves consumers adapting IDPs' interactive capabilities to fulfill traditional meanings. The second set, "crafting practices," refers to consumers proactively modifying the typical use of IDPs to achieve new meanings. The third style, "connecting practices," focuses on how consumers utilize IDPs to facilitate connections based on either the devices themselves or their shared meanings. In Study 2, we provided quantitative evidence showing that these consumer-driven value co-creation practices with IDPs positively impacted basic need satisfaction, which in turn enhanced eudaimonic well-being. And basic need satisfaction mediated the effects of consumer-IDP value co-creation practices on eudaimonic well-being. Additionally, crafting and connecting practices demonstrated stronger effects on eudaimonic well-being and basic need satisfaction than adapting practices. These findings offer new insights into the field of consumer well-being and value co-creation in smart services.

6.1. Theoretical contributions

The current research contributes to the existing literature in several ways. First, we advance understanding of the complex relationship between smart products and consumer well-being. Although innovative technological products are designed to enhance consumer well-being (Attié & Meyer-Waarden, 2022), their actual impact remains debated. Smart products can greatly improve consumer efficiency and convenience (Kang & Shao, 2023), but they may simultaneously erode consumers' sense of control and decision-making autonomy (Gonçalves et al., 2024). Our research resolves this paradox by demonstrating that consumer-driven value co-creation practices with IDPs can effectively

Table 5
Construct correlations and discriminant validity.

Variable	1	2	3	4	5	6	7
1. Adapting Practices	**0.829**						
2. Crafting Practices	0.373	**0.755**					
3. Connecting practices	0.328	0.542	**0.751**				
4. Need for Autonomy	0.344	0.574	0.616	**0.873**			
5. Need for Competence	0.348	0.571	0.607	0.468	**0.765**		
6. Need for Relatedness	0.362	0.587	0.614	0.622	0.506	**0.781**	
7. Eudaimonic Well-being	0.436	0.664	0.736	0.721	0.669	0.741	**0.830**

Note: The bold diagonal elements are square roots of AVE for each construct. Below diagonal elements are the correlations between constructs.

Table 6
Mediation analysis results of basic needs satisfaction.

Parameter	Estimate	SE	Bootstrapping (Bias-corrected 95 % CI)		
			Lower	Upper	P
Indirect effect					
AP→NFA→EW	0.020	0.011	0.004	0.049	0.018
AP→NFC→EW	0.018	0.012	0.002	0.049	0.022
AP→NFR→EW	0.024	0.013	0.004	0.055	0.017
CRP→NFA→EW	0.065	0.022	0.029	0.120	< 0.001
CRP→NFC→EW	0.053	0.022	0.019	0.106	0.002
CRP→NFR→EW	0.071	0.024	0.032	0.130	< 0.001
COP→NFA→EW	0.058	0.017	0.030	0.101	< 0.001
COP→NFC→EW	0.045	0.018	0.016	0.089	0.002
COP→NFR→EW	0.059	0.019	0.031	0.107	< 0.001
Direct effect					
AP→EW	0.062	0.029	0.009	0.123	0.022
CRP→EW	0.095	0.044	0.019	0.192	0.016
COP→EW	0.116	0.036	0.046	0.191	0.001
Total effect					
AP→EW	0.124	0.038	0.051	0.199	0.001
CRP→EW	0.284	0.055	0.184	0.404	< 0.001
COP→EW	0.277	0.043	0.200	0.367	< 0.001

enhance their eudaimonic well-being by fulfilling basic psychological needs. When consumers proactively manage their well-being by innovative technology, they not only benefit from smart services but also regain autonomy and self-efficacy. These findings align with the service research priority of "customer proactivity for well-being" (Ostrom et al., 2021; Tikkanen et al., 2023), offering a balanced perspective on how smart products genuinely enhance consumer welfare through facilitating their active participation.

Second, we expand the research on value co-creation in smart services by shifting focus to IDP contexts. Existing research has largely focused on value co-creation between consumers and independent AI products (Knote et al., 2021; Mele et al., 2022), often overlooking deeper consumer participation. In contrast, IDPs, with integrated connectivity components, substantially increase product malleability and empower consumers' dominant role in value co-creation. To bridge this gap, we investigate how consumers proactively engage with IDPs' smart and connectivity components to co-create value in daily life. We identify three styles of co-creation practices and compare their respective impacts. Our findings show that crafting and connecting practices more effectively enhance consumer eudaimonic well-being and satisfy basic psychological needs compared to adapting practices. These differences across value co-creation practices underscore the crucial role of more proactive consumer engagement in achieving deeper well-being. Thus, our research enriches the theoretical understanding of value co-creation in smart services by proposing a consumer-driven model and demonstrating its significant positive outcomes. Also, we highlight the post-adoption value creation potential of IDPs, responding to calls for greater attention to value creation in customer-owned touchpoints (Akaka & Schau, 2019).

Additionally, our research aligns with recent calls in the IS field to focus more on how newer technology can create social (non-economic) value. As Chamakiotis and Petrakaki (2025) argue, existing literature has often overlooked human agency and underemphasized how technology can foster human development. Value is frequently conceptualized narrowly as a monetized asset, neglecting its broader social dimensions. For instance, Clarke and Davison (2020) critique research on personal data markets for prioritizing corporate interests over consumer concerns, frequently focusing on helping firms minimize the cost of persuading consumers to trade privacy. In response, our research reveals that consumers can proactively manage well-being through deep engagement with digital technology. Individual well-being, grounded in user agency, forms a vital foundation for broader social welfare (Pirson, 2019). When firms design technologies and services that emphasize product malleability and empower user agency, they contribute to social welfare by enhancing individual well-being—rather than merely advancing firm-centric and economically driven value strategies. This represents a shift from transactional value extraction to socially responsible co-creation.

Lastly, we extend management studies by applying practice theory, which has increasingly been utilized in consumer behavior research (Schau et al., 2009; Thomas & Epp, 2019). Building on Shove et al. (2012) premise, there are no technical innovations without innovations in practice. We contribute to existing literature by fully accounting for material agency, particularly the interactive capabilities of IDPs, which has been underemphasized in prior work (Godfrey et al., 2022). Furthermore, we introduce the novel concepts of "connecting convergent practices" and "connecting generative practices" to clarify the nature of practice connections within the IDP context. Our work further expands the important distinction between material-based (SH devices) and meaning-based connecting practices discussed in prior literature. These findings provide new insights into the application of practice theory in management research.

6.2. Practical implications

First, our research provides valuable insights for managers aiming to enhance consumer well-being in IDP services. We found that consumer-driven value co-creation practices play a key role in addressing well-being challenges in smart services. Companies could design rewards to incentivize consumers' active engagement and encourage them to share novel IDP use cases, such as through online/offline innovation activities or user creativity-sharing platforms. Additionally, service design could focus on fulfilling consumers' basic psychological needs, ensuring advanced functionalities translate into tangible well-being benefits. By prioritizing these initiatives, firms can bridge the gap between technological advancements (e.g., IDPs) and consumer welfare.

Second, our research provides managers with a framework for fostering value co-creation practices between consumers and IDPs. It reveals that successful value co-creation requires IDPs to become seamless extensions of consumers' daily lives. Our findings have some implications for marketing and service design.

(1) In the context of IDP services, managers should further recognize consumers as active value co-creators rather than passive recipients of value. Moving beyond positioning IDPs as standalone service providers, marketing designs need to focus on

empowering consumers to fully leverage IDPs' connectivity component for personalized smart services. Brands can benefit from adopting a bottom-up logic grounded in actual consumer practices, rather than marketing pre-defined use cases. This requires reframing value propositions through a consumer lens, shifting the focus from "What can our IDPs do?" to "What can users achieve with them?" Our findings offer a roadmap for effectively communicating this evolved value narrative in the IDP marketplace.

(2) To fully unlock the potential of IDPs, managers should implement service designs that actively encourage consumer participation in value co-creation. Our research reveals the dynamic and complex nature of post-purchase consumer-IDP value co-creation, emphasizing the importance of supporting consumers, especially non-technical users, in developing adapting, crafting, and connecting practices. Practical implementations can include: (a) developing modular smart scene templates to simplify setup processes, (b) optimizing user interfaces with clear technical guidance and real-world use cases to reduce cross-device complexity, and (c) establishing ongoing support mechanisms, such as seamless integration of new devices, experimentation incentives, and feedback-driven function updates. Through these service designs, companies can help consumers continuously enhance their usage experience and fully realize the potential for value co-creation with IDPs.

Third, our research offers useful suggestions for consumers on how to proactively manage their well-being through digital technologies. Our findings highlight the importance of consumers taking on the role of active prosumers rather than passive recipients of smart services. We recommend that consumers begin with basic device linkages, gradually master advanced programming skills for personalized smart scenarios, and engage family members in optimizing their SH usages. These actionable strategies are designed to boost both confidence and competence in using digital tools for well-being management. Through this process, consumers can elevate IDPs from simple functional devices to meaningful enablers of long-term personal well-being, embracing the role of proactive well-being managers.

6.3. Limitations and directions for future research

This research has several limitations that should be taken into consideration in the future. First, we selected SH as the study context. Although we have interpreted four reasons for this choice, whether our findings can be generalized to other types of IDPs needs to be further examined. Future studies should explore value co-creation practices in other IDP contexts. Second, our research does not pay attentions on particular user groups, such as residents of high-end or villa-level SH. Future research could further compare how value co-creation practices and their effects may vary across different user groups. Third, we identified three types of consumer-IDP value co-creation practices, representing a new attempt in the smart service field. While we followed standard procedures to develop corresponding measurement scales, there remain limitations in item construction. Further research is needed to refine these scales and conduct empirical studies to better clarify their conceptual connotations and explore outcomes. Fourth, our research compared the effects of different value co-creation practices on eudaimonic well-being and basic psychological need satisfaction. Future work could test additional comparative hypotheses, such as how certain practices may differentially influence three basic psychological needs. Finally, given the rapid evolution of digital technologies, resolving the well-being controversy in smart services is increasingly urgent. While we advocate a consumer-driven value co-creation perspective, future research should also explore alternative frameworks.

7. Conclusion

Our research offers new insights into the paradox between smart products and consumer well-being. We propose resolving the well-being paradox in smart services through consumer-driven value co-creation with IDPs. Specifically, we identify three consumer-IDP value co-creation practices: adapting practices, crafting practices, and connecting practices, which positively impact eudaimonic well-being. We empirically examine the positive effects of these practices on eudaimonic well-being by satisfying consumers' basic psychological needs. Notably, crafting and connecting practices have stronger effects on both eudaimonic well-being and need satisfaction compared to adapting practices. Importantly, our findings highlight the vital role of IDPs' connectivity components in empowering consumer-driven value co-creation. We reveal how consumers leverage IDPs to proactively manage their well-being. These insights not only expand the theoretical understanding of consumer well-being and value co-creation in IDP services, but also provide practical guidance for brands on facilitating consumer-driven value co-creation and enhancing consumer well-being.

CRediT authorship contribution statement

Jun Gao: Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Shuying Gong:** Writing – original draft, Supervision. **Huichao Guo:** Writing – review & editing, Project administration, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijinfomgt.2025.102979.

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